Unit 3 · Outcome 1 · Maintaining Biodiversity

Threats to Biodiversity

KK10 VCE Environmental Science · Units 3 & 4 · Exam study summary

> **Key knowledge.** 10 Why do species slip toward extinction? A species rarely vanishes for a single reason

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

Why do species slip toward extinction?

Kestrel Ridge Reserve — a fragmented riparian swamp where the last wild Helmeted Honeyeaters persist. Drag to orbit. Key Knowledge 10 Why do species slip toward extinction? A species rarely vanishes for a single reason. It is squeezed by a web of pressures — some caused by people, some not — that interact and compound. KK10 is about naming those pressures precisely and explaining the mechanism by which each one reduces a population's size, range or genetic health. VCAA dot point: human and non-human threats to biodiversity — creation and isolation of sma

The threat map

Orientation The threat map Eight pressures sit behind the VCAA dot point. Learn them as a set, tag each as predominantly human or non-human, and — crucially — learn the one-line mechanism for each. Hover a card; the mechanism is what earns marks. Habitat loss human Clearing, draining or fragmenting habitat removes the resources a species needs and splits one large population into small isolated ones. Over-exploitation human Harvesting, hunting or collecting faster than a population can replace itself drives numbers down and shrinks range. Inb

Habitat loss & the small-population trap

⚠ **EXAM TRAPS**

Threat 1 & 2 Habitat loss & the small-population trap Habitat loss and over-exploitation are the two engines that create small, isolated populations — and small isolated populations are where most other threats do their damage. Use the explorer below to watch Kestrel Ridge fragment and the honeyeater population thin out. From loss to fragmentation to isolation Three related ideas often get muddled. Keep them distinct: Term What happens Why it threatens the species Habitat loss Total area of suitable habitat falls (e.g. swamp drained for farmland). Fewer

Term	What happens	Why it threatens the species
Habitat loss	Total area of suitable habitat falls (e.g. swamp drained for farmland).	Fewer resources → the area can support fewer individuals (lower carrying capacity).
Fragmentation	Remaining habitat is broken into disconnected patches.	Patches are too small to be self-sustaining; movement between them is risky.
Isolation	Sub-populations can no longer interbreed across patches.	No gene flow → each patch drifts genetically and is vulnerable to local extinction.

Inbreeding & the extinction vortex

Threat 3 Inbreeding & the extinction vortex Once a population is small, a genetic trap snaps shut. Inbreeding raises the chance that offspring inherit two copies of a harmful recessive allele, while genetic diversity — the raw material for adapting to change — drains away. Watch it happen in the bottleneck simulator. Scene B · Genetic Bottleneck Shrink the population and breed generations — watch allele variety collapse. Population 120 Generations 0 Allele types 8 Diversity 100% Surviving population 120 ▶ Breed one generation ↺ gene pool ◆ Two distinct

Lost partners — pollinators, dispersers, hosts & symbionts

Threat 4 Lost partners — pollinators, dispersers, hosts & symbionts No species lives alone. Many depend on a partner to reproduce or persist. The VCAA wording is specific: loss of pollinators, dispersal agents, host species or symbionts that affect reproduction and persistence. Lose the partner and you threaten the species — even if nothing touches it directly. Pollinators Insects, birds or bats that move pollen so plants can set seed. No pollinator → no seed → no next generation of plants (and the animals that feed on them). Dispersal agents Animals

Bioaccumulation & biomagnification

Threat 5 · high-frequency exam topic Bioaccumulation & biomagnification This is the most mixed-up pair of terms in the whole course — and examiners love testing whether you can tell them apart. Build the tower below and watch a persistent pollutant climb a food chain. ◆ Bioaccumulation Build-up of a persistent pollutant within a single organism over its lifetime, because it is absorbed faster than it can be broken down or excreted. (Think: one fish, getting more contaminated as it ages.) ◆ Biomagnification Increase in pollutant concentration from one tro

Feature	Bioaccumulation	Biomagnification
Where	Within one organism	Between trophic levels
Over what	The organism's lifetime	Along a food chain
Requires	A persistent, fat-soluble toxin	Bioaccumulation + a food chain
Worst hit	Long-lived individuals	Top predators (e.g. the honeyeater, raptors)

Climate change, disease & introduced species

Threats 6 · 7 · 8 Climate change, disease & introduced species The final three threats often act as multipliers — they hit hardest when a population is already small, fragmented or genetically uniform from the threats above. Climate change Shifting temperature and rainfall move the climate envelope a species is adapted to. Species must shift range, adapt, or decline — and a fragmented landscape blocks the shift. For Kestrel Ridge, hotter, drier summers raise bushfire risk (the 2009 fires came within kilometres of Yellingbo) and dry out the swamp the h

Invader impact	Mechanism	Example
Competition	Takes shelter, food or water natives rely on	Rabbits stripping ground cover; miners excluding small birds
Predation	Eats natives that have no evolved defence	Foxes & cats on ground-nesting birds
Disease / parasites	Introduces pathogens natives can't resist	Chytrid fungus in frogs
Habitat change	Alters the habitat structure itself	Carp muddying waterways; weeds choking swamp

Sort it — human vs non-human

◇ **ACTIVE RECALL**

Interactive · check yourself Sort it — human vs non-human Drag each scenario into the bin where its primary driver belongs. Some are deliberately tricky (climate change and disease can be both — decide based on how the scenario is framed). Then hit Check . Primarily human 0 Primarily non-human 0 ✓ s ↺

Build a causal chain

Build skill · cause → mechanism → effect Build a causal chain Top-band answers always trace a chain : a named threat, the mechanism it works through, and the measurable effect on the species. Pick a threat below to see a worked chain, then try writing your own in the Writing Lab. Habitat loss Inbreeding Biomagnification Introduced species ✦ Why this works

Command words, P·E·L & exam traps

⚠ **EXAM TRAPS**

Writing lab Command words, P·E·L & exam traps Knowing the content isn't enough — you must answer the verb . Tap a command word to see what the examiner actually wants, then drill the P·E·L structure on a real prompt. Decode the command word P·E·L writing trainer A reliable structure for short-answer extended responses: Point (state it), Explain (the mechanism), Link (back to the species/question). Draft each line, then reveal a model. Explain how habitat fragmentation threatens the long-term survival of the Helmeted Honeyeater. (3 marks) P Point Name fra

Five VCAA-style questions

★ **PRACTICE & SELF-MARK**

Practice exam · self-mark Five VCAA-style questions Attempt each question before revealing the responses. Compare the weak and strong answers, read the marking guide, then mark yourself honestly using the buttons. Your total tracks in the dock and saves automatically.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.